GENESIS
YOUR POTENTIAL. OUR PASSION

Full Stack Course Curriculum

PYTHON

PYTHON INTRO
- OVERVIEW OF PYTHON PROGRAMMING LANGUAGE AND ITS FEATURES

PYTHON COMMENTS
- SINGLE-LINE COMMENTS
- MULTI-LINE COMMENTS

PYTHON VARIABLES
- VARIABLE DECLARATION
- VARIABLE NAMING RULES
- VARIABLE ASSIGNMENT

PYTHON DATA TYPES
- NUMERIC DATA TYPES (INT, FLOAT, COMPLEX)
- TEXT DATA TYPE (STRING)
- BOOLEAN DATA TYPE
- SEQUENCE DATA TYPES (LIST, TUPLE)
- SET DATA TYPE
- MAPPING DATA TYPE (DICTIONARY)

PYTHON STRINGS
- CREATING STRINGS
- STRING INDEXING AND SLICING
- STRING METHODS AND OPERATIONS

PYTHON BOOLEANS
- TRUE AND FALSE VALUES
- BOOLEAN OPERATORS (AND, OR, NOT)

PYTHON OPERATORS
- ARITHMETIC OPERATORS
- COMPARISON OPERATORS
- LOGICAL OPERATORS
- ASSIGNMENT OPERATORS
- IDENTITY OPERATORS
- MEMBERSHIP OPERATORS

PYTHON LISTS
- CREATING LISTS

- ACCESSING LIST ELEMENTS
- MODIFYING LISTS
- LIST METHODS AND OPERATIONS

PYTHON TUPLES

- CREATING TUPLES
- ACCESSING TUPLE ELEMENTS
- MODIFYING TUPLES
- TUPLE METHODS AND OPERATIONS

PYTHON SETS

- CREATING SETS
- ACCESSING SET ELEMENTS
- MODIFYING SETS
- SET METHODS AND OPERATIONS

PYTHON DICTIONARIES

- CREATING DICTIONARIES
- ACCESSING DICTIONARY ELEMENTS
- MODIFYING DICTIONARIES
- DICTIONARY METHODS AND OPERATIONS

PYTHON IF...ELSE

- IF STATEMENTS
- IF...ELSE STATEMENTS
- IF...ELIF...ELSE STATEMENTS
- NESTED IF STATEMENTS

PYTHON WHILE LOOPS

- WHILE LOOP SYNTAX
- BREAK AND CONTINUE STATEMENTS
- NESTED WHILE LOOPS

PYTHON FOR LOOPS

- FOR LOOP SYNTAX
- ITERATING OVER SEQUENCES AND RANGES
- BREAK AND CONTINUE STATEMENTS
- NESTED FOR LOOPS

PYTHON FUNCTIONS

- DEFINING FUNCTIONS
- FUNCTION ARGUMENTS (POSITIONAL, DEFAULT, KEYWORD)
- RETURN STATEMENT
- FUNCTION SCOPE

PYTHON LAMBDA

- LAMBDA FUNCTIONS (ANONYMOUS FUNCTIONS)
- SYNTAX AND USAGE

PYTHON ARRAYS

- INTRODUCTION TO ARRAYS IN PYTHON
- WORKING WITH ARRAYS USING NUMPY LIBRARY

PYTHON CLASSES/OBJECTS

- INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING (OOP)
- CLASS DEFINITION
- CREATING OBJECTS (INSTANCES)
- CLASS ATTRIBUTES AND METHODS

PYTHON ABSTRACTIONS

- ENCAPSULATION
- ABSTRACTION USING CLASSES AND OBJECTS

PYTHON INHERITANCE

- INHERITING CLASSES
- OVERRIDING METHODS
- INHERITANCE TYPES (SINGLE, MULTIPLE, MULTI-LEVEL)

PYTHON POLYMORPHISM

- POLYMORPHISM CONCEPTS
- METHOD OVERRIDING
- METHOD OVERLOADING

PYTHON TRY...EXCEPT

- EXCEPTION HANDLING IN PYTHON
- TRY...EXCEPT BLOCK
- HANDLING SPECIFIC EXCEPTIONS
- FINALLY BLOCK

FILE HANDLING

- OVERVIEW OF FILE HANDLING IN PYTHON
- READING, WRITING, AND MANIPULATING FILES

PYTHON READ FILES

- OPENING AND CLOSING FILES
- READING FILE CONTENTS
- READING LINE BY LINE
- READING SPECIFIC PORTIONS OF A FILE

PYTHON WRITE/CREATE FILES

- CREATING AND WRITING TO FILES
- APPENDING TO EXISTING FILES
- WRITING DATA IN DIFFERENT FORMATS (TEXT, CSV, ETC.)

PYTHON DELETE FILES

- DELETING FILES USING PYTHON
- ERROR HANDLING AND PERMISSIONS

SQL

SQL INTRO
SQL SYNTAX
SQL DATA TYPES
SQL OPERATORS

SQL DATABASE

- SQL CREATE DATABASE
- SQL DROP DATABASE
- SQL RENAME DATABASE
- SQL SELECT DATABASE

SQL TABLE

WHAT IS TABLE

- SQL CREATE TABLE
- SQL DROP TABLE SQL
- DELETE TABLE SQL
- RENAME TABLE SQL
- TRUNCATE TABLE SQL
- COPY TABLE SQL
- TEMP TABLE SQL
- ALTER TABLE

SQL SELECT

- SELECT STATEMENT
- SQL SELECT UNIQUE
- SQL SELECT DISTINCT
- SQL SELECT COUNT
- SQL SELECT TOP
- SQL SELECT FIRST
- SQL SELECT LAST
- SQL SELECT RANDOM
- SQL SELECT IN

- SQL SELECT MULTIPLE
- SQL SELECT DATE
- SQL SELECT SUM
- SQL SELECT NULL

SQL CLAUSE

- SQL WHERE
- SQL AND
- SQL OR
- SQL WITH
- SQL AS
- SQL HAVING CLAUSE

SQL ORDER BY

- ORDER BY CLAUSE
- ORDER BY ASC
- ORDER BY DESC
- ORDER BY RANDOM
- ORDER BY LIMIT
- ORDER BY MULTIPLE COLS

SQL INSERT

- INSERT STATEMENT
- INSERT MULTIPLE ROWS

SQL UPDATE

- UPDATE STATEMENT
- SQL UPDATE JOIN
- SQL UPDATE DATE

SQL DELETE

- DELETE STATEMENT
- SQL DELETE TABLE
- SQL DELETE ROW
- SQL DELETE ALL ROWS
- DELETE DUPLICATE ROWS
- SQL DELETE DATABASE
- SQL DELETE VIEW
- SQL DELETE JOIN

SQL JOIN

- SQL OUTER JOIN
- SQL LEFT JOIN
- SQL RIGHT JOIN
- SQL FULL JOIN
- SQL CROSS JOIN

SQL KEYS

- PRIMARY KEY
- FOREIGN KEY
- COMPOSITE KEY
- UNIQUE KEY
- ALTERNATE KEY

DIFFERENCE

- SQL VS NO
- SQLIN VS EXISTS
- GROUP BY VS ORDER BY
- WHERE VS HAVING
- WHERE CONDITION IN SQL

SQL INJECTION

- SQL INJECTION

SQL STRING FUNCTIONS

- SQL STRING FUNCTIONS
- LTRIM FUNCTION IN SQL
- UPPER FUNCTION IN SQL
- RTRIM FUNCTION IN SQL

DJANGO

DJANGO TUTORIAL

- INTRODUCTION TO DJANGO WEB FRAMEWORK
- OVERVIEW OF ITS FEATURES AND BENEFITS

DJANGO INSTALLATION

- INSTALLING DJANGO FRAMEWORK ON YOUR LOCAL MACHINE

DJANGO PROJECT

- CREATING A NEW DJANGO PROJECT
- PROJECT STRUCTURE AND CONFIGURATION FILES

APACHE CONFIGURATION

- CONFIGURING APACHE WEB SERVER FOR DJANGO PROJECT DEPLOYMENT

VIRTUAL ENVIRONMENT SETUP

- SETTING UP A VIRTUAL ENVIRONMENT FOR DJANGO PROJECTS

ADMIN INTERFACE

- USING DJANGO'S BUILT-IN ADMIN INTERFACE FOR SITE ADMINISTRATION

DJANGO APP

- CREATING A NEW DJANGO APP WITHIN A PROJECT
- APP STRUCTURE AND FILES

DJANGO MVT

- UNDERSTANDING THE MODEL-VIEW-TEMPLATE (MVT) ARCHITECTURAL PATTERN IN DJANGO

DJANGO MODEL

- CREATING MODELS TO DEFINE DATABASE TABLES AND RELATIONSHIPS

DJANGO VIEW

- WRITING VIEWS TO HANDLE HTTP REQUESTS AND RESPONSES

DJANGO TEMPLATE

- CREATING HTML TEMPLATES TO DEFINE THE PRESENTATION LAYER

URL MAPPING

- MAPPING URLS TO CORRESPONDING VIEWS IN DJANGO

STATIC FILES HANDLING

- MANAGING AND SERVING STATIC FILES (CSS, JAVASCRIPT, IMAGES) IN DJANGO

MODELFORMS

- CREATING FORMS FROM DJANGO MODELS USING MODELFORMS

DJANGO FORMS

- BUILDING AND HANDLING FORMS IN DJANGO

FORM VALIDATION

- VALIDATING FORM DATA IN DJANGO

FILE UPLOAD

- HANDLING FILE UPLOADS IN DJANGO

DATABASE CONNECTIVITY

- CONNECTING DJANGO TO A DATABASE (SQLITE, MYSQL, POSTGRESQL)

DATABASE MIGRATIONS

- PERFORMING DATABASE MIGRATIONS WITH DJANGO

DJANGO MIDDLEWARE

- INTRODUCTION TO DJANGO MIDDLEWARE AND ITS USAGE

REQUEST AND RESPONSE

- WORKING WITH HTTP REQUESTS AND RESPONSES IN DJANGO

DJANGO EXCEPTIONS

- HANDLING EXCEPTIONS AND ERRORS IN DJANGO

DJANGO SESSION

- MANAGING USER SESSIONS IN DJANGO

DJANGO COOKIE

- WORKING WITH COOKIES IN DJANGO

DJANGO CSV OUTPUT

- GENERATING CSV OUTPUT IN DJANGO

DJANGO PDF OUTPUT

- GENERATING PDF OUTPUT IN DJANGO

DJANGO AND BOOTSTRAP

- INTEGRATING BOOTSTRAP FRAMEWORK WITH DJANGO FOR STYLING

DEPLOY ON GITHUB

- DEPLOYING A DJANGO PROJECT ON GITHUB

DJANGO MAIL SETUP

- CONFIGURING EMAIL SENDING IN DJANGO

DJANGO DEFAULT CRUD

- OVERVIEW OF DJANGO'S BUILT-IN CRUD (CREATE, READ, UPDATE, DELETE) OPERATIONS

CRUD APPLICATION

- BUILDING A CRUD APPLICATION IN DJANGO

DJANGO REDIRECTS

- REDIRECTING USERS TO DIFFERENT URLS IN DJANGO

DJANGO ON_DELETE

- UNDERSTANDING THE ON_DELETE BEHAVIOR FOR RELATED MODELS IN DJANGO

DJANGO CLASS-BASED GENERIC VIEWS

- WORKING WITH DJANGO'S CLASS-BASED GENERIC VIEWS FOR COMMON TASKS

DJANGO USERCREATIONFORM

- USING DJANGO'S USERCREATIONFORM FOR USER REGISTRATION

DJANGO IMAGE UPLOAD

- HANDLING IMAGE UPLOADS IN DJANGO

DJANGO ORM QUERIES

- PERFORMING DATABASE QUERIES USING DJANGO'S ORM (OBJECT-RELATIONAL MAPPING)

HTML

INTRODUCTION TO HTML

- OVERVIEW OF HTML (HYPERTEXT MARKUP LANGUAGE)
- ROLE OF HTML IN CREATING WEB PAGES

HTML TAGS

- INTRODUCTION TO HTML TAGS AND THEIR SYNTAX
- COMMONLY USED TAGS IN HTML

HTML ATTRIBUTES

- UNDERSTANDING HTML ATTRIBUTES AND THEIR USAGE
- ADDING ATTRIBUTES TO HTML TAGS

HTML EDITORS

- POPULAR HTML EDITORS AND DEVELOPMENT ENVIRONMENTS
- CHOOSING THE RIGHT HTML EDITOR FOR YOUR NEEDS

HTML DOCUMENT STRUCTURE

- STRUCTURE AND LAYOUT OF AN HTML DOCUMENT
- HTML DOCTYPE DECLARATION AND BASIC STRUCTURE

HTML HEADINGS

- USING HEADINGS TO STRUCTURE CONTENT IN HTML
- DIFFERENT LEVELS OF HEADINGS (H1 TO H6)

HTML PARAGRAPHS

- CREATING AND FORMATTING PARAGRAPHS IN HTML
- USING PARAGRAPH TAGS FOR TEXT CONTENT

HTML STYLES

- ADDING INLINE STYLES TO HTML ELEMENTS
- USING CSS FOR STYLING HTML CONTENT

HTML COMMENTS

- ADDING COMMENTS TO HTML CODE FOR DOCUMENTATION PURPOSES
- SYNTAX AND USAGE OF HTML COMMENTS

HTML BLOCKS, CLASSES, IDS

- UNDERSTANDING BLOCK-LEVEL AND INLINE ELEMENTS IN HTML
- ASSIGNING CLASSES AND IDS TO HTML ELEMENTS FOR STYLING AND JAVASCRIPT PURPOSES

HTML TABLES

- CREATING TABLES IN HTML
- DEFINING TABLE ROWS, COLUMNS, AND HEADERS

HTML LISTS

- CREATING ORDERED AND UNORDERED LISTS IN HTML
- USING LIST TAGS (UL, OL, LI) FOR LIST ITEMS

HTML IMAGES

- INSERTING AND DISPLAYING IMAGES IN HTML
- ADDING IMAGE ATTRIBUTES (ALT, SRC) FOR ACCESSIBILITY

HTML FORMS

- BUILDING FORMS IN HTML FOR USER INPUT AND DATA SUBMISSION
- FORM CONTROLS AND FORM ATTRIBUTES

HTML FILE PATHS

- UNDERSTANDING FILE PATHS IN HTML
- RELATIVE AND ABSOLUTE FILE PATHS FOR LINKING RESOURCES (IMAGES, STYLESHEETS, SCRIPTS)

CSS

INTRODUCTION TO CSS

- OVERVIEW OF CSS (CASCADING STYLE SHEETS)
- ROLE OF CSS IN STYLING HTML ELEMENTS

CSS SELECTORS

- UNDERSTANDING CSS SELECTORS AND THEIR USAGE
- DIFFERENT TYPES OF SELECTORS (ELEMENT, CLASS, ID, ATTRIBUTE, ETC.)

INSERTING CSS INTO HTML

- INLINE CSS: ADDING CSS STYLES DIRECTLY TO HTML ELEMENTS
- INTERNAL CSS: ADDING CSS STYLES WITHIN THE <STYLE> TAGS IN THE HTML FILE
- EXTERNAL CSS: LINKING AN EXTERNAL CSS FILE TO THE HTML FILE

COLORS, BORDERS, BACKGROUND, MARGINS, PADDING

- APPLYING COLORS TO HTML ELEMENTS USING CSS
- STYLING BORDERS, BACKGROUNDS, MARGINS, AND PADDING USING CSS PROPERTIES

WORKING WITH CORE CONCEPTS OF CSS

- BOX MODEL: UNDERSTANDING THE CONCEPT OF BOX MODEL IN CSS
- OUTLINE: ADDING OUTLINE STYLES TO ELEMENTS
- FONTS: SPECIFYING FONT STYLES, SIZES, AND FAMILIES
- LINKS: STYLING HYPERLINKS WITH CSS
- LISTS: CUSTOMIZING THE APPEARANCE OF ORDERED AND UNORDERED LISTS
- TABLES: STYLING TABLES USING CSS PROPERTIES
- COMBINATIONS: APPLYING CSS PROPERTIES AND STYLES IN COMBINATION

Full Stack Python

JAVASCRIPT

INTRODUCTION TO JAVASCRIPT

- OVERVIEW OF JAVASCRIPT AS A SCRIPTING LANGUAGE
- USAGE OF SCRIPTING LANGUAGES IN WEB DEVELOPMENT

GETTING STARTED WITH JAVASCRIPT

- SETTING UP AND GETTING STARTED WITH JAVASCRIPT DEVELOPMENT

HISTORY OF JAVASCRIPT

- BRIEF HISTORY AND EVOLUTION OF JAVASCRIPT AS A PROGRAMMING LANGUAGE

VARIABLES

- DECLARING AND WORKING WITH VARIABLES IN JAVASCRIPT
- VARIABLE SCOPES AND HOISTING

DATA TYPES

- OVERVIEW OF DATA TYPES IN JAVASCRIPT
- PRIMITIVE AND COMPLEX DATA TYPES

REAL-TIME EXAMPLES

- EXAMPLES ILLUSTRATING THE USAGE OF JAVASCRIPT IN REAL-WORLD SCENARIOS

OPERATORS AND CONTROL STATEMENTS

- TYPES OF OPERATORS IN JAVASCRIPT AND THEIR USAGE
- CONDITIONAL CONTROL STATEMENTS (IF, IF-ELSE, IF-ELSE IF)
- REAL-TIME EXAMPLES SHOWCASING CONDITIONAL CONTROL STATEMENTS

LOOPS

- WHILE LOOP AND ITS USAGE
- DO-WHILE LOOP AND ITS USAGE
- FOR LOOP AND ITS USAGE
- KEYWORDS USED IN CONTROL STATEMENTS LIKE RETURN AND BREAK
- REAL-TIME EXAMPLES OF LOOP USAGE

WORKING WITH FUNCTIONS

- DEFINING FUNCTIONS IN JAVASCRIPT
- FUNCTION EXPRESSIONS AND ANONYMOUS FUNCTIONS
- CALLING FUNCTIONS AND PASSING ARGUMENTS

- FUNCTION SCOPE AND CLOSURES
- NESTED FUNCTIONS
- FIRST-CLASS FUNCTIONS AND HIGHER-ORDER FUNCTIONS
- IMMEDIATELY INVOKED FUNCTION EXPRESSIONS (IIFE)
- REAL-TIME EXAMPLES DEMONSTRATING FUNCTION USAGE

WORKING WITH ARRAYS

- CREATING AND MANIPULATING ARRAYS IN JAVASCRIPT
- ACCESSING, LOOPING OVER, AND REMOVING ELEMENTS FROM ARRAYS
- ARRAY METHODS IN DETAIL
- REAL-TIME EXAMPLES ILLUSTRATING ARRAY OPERATIONS

WORKING WITH NUMBERS AND DATES

- WORKING WITH NUMBERS IN JAVASCRIPT
- NUMBER METHODS AND OPERATIONS
- WORKING WITH DATES AND TIME IN JAVASCRIPT
- DATE METHODS AND OPERATIONS
- REAL-TIME EXAMPLES INVOLVING NUMBERS AND DATES

WORKING WITH STRINGS

- MANIPULATING STRINGS IN JAVASCRIPT
- STRING METHODS AND OPERATIONS
- REAL-TIME EXAMPLES SHOWCASING STRING OPERATIONS

WORKING WITH OBJECTS

- OBJECTS AND PROPERTIES IN JAVASCRIPT
- CREATING NEW OBJECTS
- CREATING OBJECTS USING CONSTRUCTOR FUNCTIONS AND THE **CREATE** METHOD
- UNDERSTANDING THE **THIS** KEYWORD
- REAL-TIME EXAMPLES INVOLVING OBJECTS AND PROPERTIES

WORKING WITH SCOPES AND EVENTS

- TYPES OF SCOPES IN JAVASCRIPT (GLOBAL SCOPE, FUNCTION SCOPE, BLOCK SCOPE)
- DIFFERENT TYPES OF EVENTS IN JAVASCRIPT
- REAL-TIME EXAMPLES DEMONSTRATING EVENT HANDLING

WORKING WITH VALIDATIONS

- BASIC FORM VALIDATIONS USING JAVASCRIPT
- ADVANCED FORM VALIDATIONS USING REGULAR EXPRESSIONS
- REAL-TIME EXAMPLES SHOWCASING FORM VALIDATIONS

WORKING WITH THE DOCUMENT OBJECT MODEL (DOM)

- INTRODUCTION TO THE DOCUMENT OBJECT MODEL (DOM)
- WORKING WITH DOM METHODS, DOCUMENTS, AND ELEMENTS
- HANDLING EVENTS IN JAVASCRIPT
- ADDING EVENT LISTENERS TO DOM ELEMENTS

WORKING WITH DOM NAVIGATION AND NODES

- NAVIGATING THE DOM TREE STRUCTURE
- WORKING WITH DOM NODES AND COLLECTIONS

- PROJECT (INCLUDES BOTH FRONTEND & BACKEND).

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